# Switching the loop on at inference cuts E rate ≈15×

exp038 · 30 May 2026

The trained networks this entry uses are produced once in the shared training hub, exp022 (Training), and reused here rather than retrained.

## Abstract

Loads a trained COBA network and switches the I-loop on at inference by sweeping *ei_strength* from 0 to 1. Across independently trained seeds 42–44, the same feedforward weights that fire at ≈ 133 Hz without the loop fire at ≈ 9 Hz with it engaged, a ≈ 15× drop with no weight update. Accuracy, though, falls ≈ 36 pp (from ≈ 90% to ≈ 55%) as the loop engages, because the readout was never trained with it. PING gating is a post-hoc sparsity knob: the architecture, not the training, supplies the gamma dynamics, but using it well still needs training *with* the loop.

## Method

Each inference intervention starts from the checkpoint selected by validation accuracy. The recurrent-loop replacements and input-rate sweeps therefore probe the selected classifier, not the final-epoch training state.

| Parameter | Value |
|---|---|
| Integration timestep $\Delta t$ | 0.1 ms |
| Trial duration $T$ | 200 ms |
| Evaluation corpus | Official MNIST test partition (10000 samples) |
| Evaluated held-out samples per seed | 14000 |
| Epochs | 50 |

The PING and COBA baseline definitions and the training recipe are in exp025 this entry runs the COBA → PING I-loop transfer probe at eval time on the trained baselines. (The input-rate sweep / f–I curve material that previously lived here has moved to exp023, the natural home for "architectural response to drive".)

**Inference-time probe.** The trained COBA baselines (seeds 42–44, activity regulariser off) are loaded and *ei_strength* (the I-loop gain) is overridden at eval time across 11 values from 0 to 1. Each seed uses the same grid and held-out sample count. $W_{\text{in}}$ and $W_{\text{out}}$ load from its COBA checkpoint; $W^{EI}$ and $W^{IE}$ are freshly initialised (the COBA checkpoint stores these at zero, so skipping the load leaves a functional I-loop). No retraining. The quantitative curves report the across-seed mean and shaded sample SD. The raster panels show seed 42 as an explicitly illustrative example.

## Results

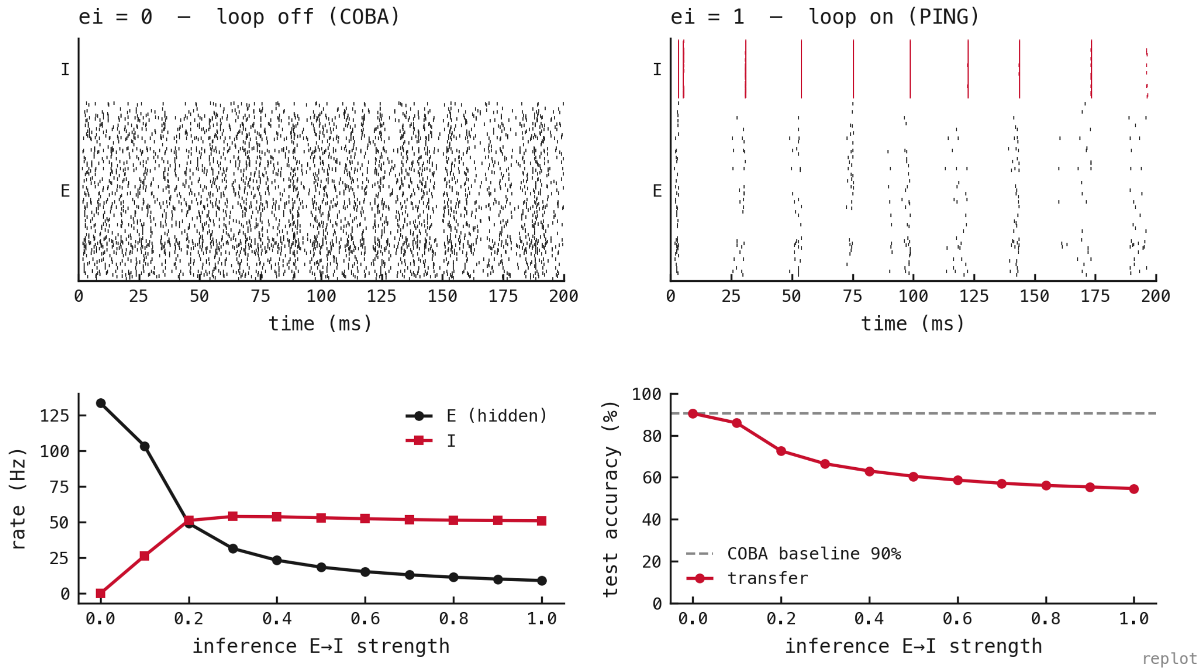


Figure 1: Switching the recurrent I-loop on *at inference* on a trained COBA network (no retraining; $W^{EI}/W^{IE}$ freshly wired, since the COBA checkpoint stores them at zero). **Top**: the same feedforward weights fire densely and asynchronously at $ei = 0$ (COBA) and in gamma bands at $ei = 1$ (PING); the gamma dynamics come from the inhibitory architecture, not from training. **Bottom left**: across-seed mean E rate falls ≈ 15× (≈ 133 → 9 Hz) as the loop engages while I rises to ≈ 51 Hz; the suppression is continuous in loop strength. **Bottom right**: mean accuracy *degrades* without retraining, from the ≈ 90% COBA baseline to ≈ 55% at full strength (a ≈ 36 pp cost). Shaded bands show sample SD across seeds 42–44. So the architecture supplies the rate-gating for free, but using it well needs training *with* the loop (exp025): the sparsity is architectural, the accuracy is learned.

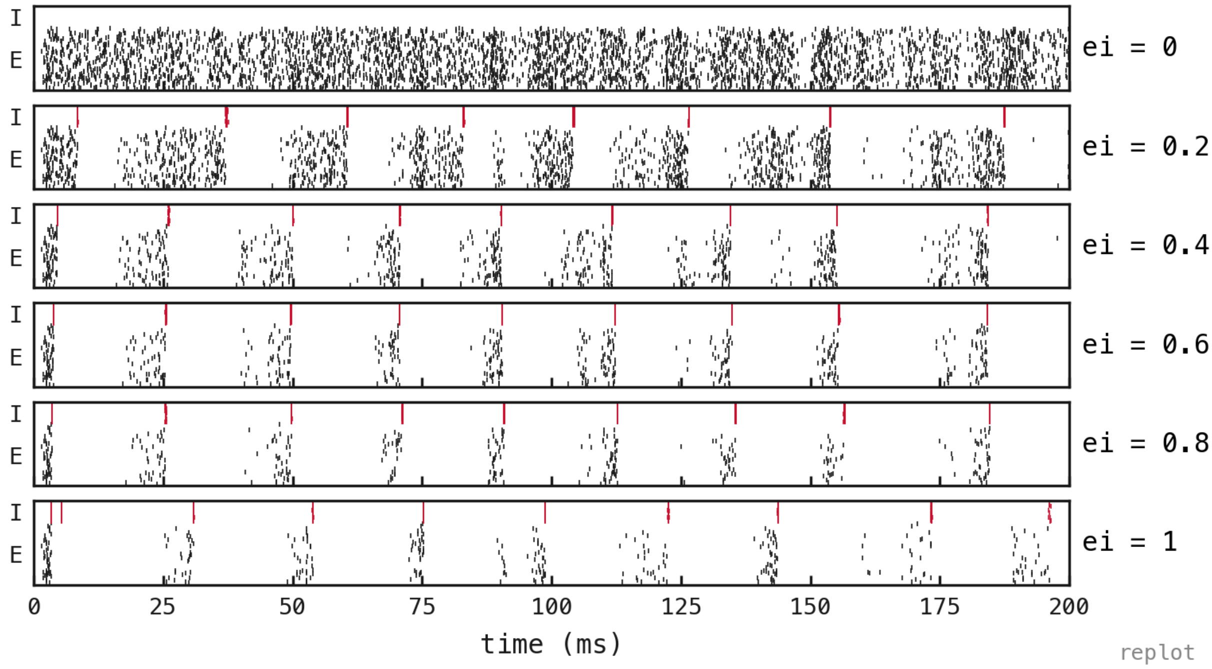


Figure 2: The illustrative seed-42 transition: trained COBA replayed at six inference-time *ei_strength* values (same trial, same feedforward weights, a fresh I-loop each row). At $ei = 0$ the asynchronous-dense COBA pattern persists; by $ei \approx 0.4$ the same weights produce gamma cycles, sharpening toward $ei = 1$. The rhythm appears continuously as the loop is wired in, with no retraining at any point.